

Subject Code	AF5634
Subject Title	Economics of Sustainability
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	None
Objectives	The subject is one on economics of the environment, with special attention on sustainability. This subject aims to provide an integrated overview of environmental economics and sustainable development from a global perspective. Students are introduced to the economic concepts and theories for analysing sustainable development and practices. It also focuses on the policy instruments which have been suggested as a means of achieving sustainability. It also investigates the impact of policy instruments on economics with the application of data analysis.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - demonstrate deeper knowledge and understanding of mainstream economics and alternative economic paradigms of sustainability - provide informed comment on the interconnections between the natural environment and the operations of our economic system - understand the role of microeconomics and macroeconomics in sustainable development - demonstrate in-depth knowledge of different economic tools for assessing environmental problems - demonstrate in-depth knowledge of different strategies for reconciling environmental concerns with requirements of economic growth and development - demonstrate awareness of the ethical implications and considerations involved in the economic approach to environmental problems - understand how economists make use of data to analyse the impact of climate-change policies
Subject Synopsis/ Indicative Syllabus	Economic Theory and Sustainable Development Economics and the environment Fundamental concepts in environmental economics Identifying environmental objectives – Sustainable development

	Concepts of sustainability Trends and challenges of sustainability Environmental policy planning Measuring and Achieving Sustainable Development: Macroeconomic Approaches Defining conventional gross net product (GNP) Modifying GNP for missing (nonmarket) values – green GNP Genuine savings Critical capital concerns and strong sustainability Monetary and fiscal policies Modeling the Market Process: A Review of the Basics Market models: The Fundamentals The Model of Supply and Demand Economic criteria of efficiency Welfare measures Modeling Market Failure Environmental problems: A market failure Externalities Modeling environmental damage as negative externality The absence of property rights Conventional Solutions to Environmental Problems The Command-and-Control Strategies: The Case of Standards Incentive-Based Strategies: Emission Charges and Subsidies Incentive-Based Strategies: Transferable Discharge Permits Usage of data to analyse the impact of climate-change policies Analytical Tools for Environmental Planning Concept of risk Risk assessment Risk management Assessing costs and benefits for environmental decision making Benefit-cost analysis in environmental decision making Ethics and Environmental Economics The ethical limits of environmental economics Ethical concerns of economic value-judgment
Teaching/Learning Methodology	The course is comprised of a seminar of three hours per week, students' presentations, and Individual assignments. The 3-hour seminar per week includes a 2-hour lecture and 1-hour tutorial. The lectures will be structured to help students to understand various topics relating to environment economics and sustainability. The tutorials will provide students with the opportunity to deepen their understanding of the concepts taught in lectures and to apply the theories and economic tools to the analysis of environmental problems. The activities in tutorials include presentation and discussion of tutorial questions and environmental case studies.

	Students are required to submit individual assignments of different topics and they are expected to interact with their lecturer and other classmates in preparing answers for the questions in the assignments. To reflect the significant technology content in this subject, 10% (or more) of the overall weighting of this subject is based on individual assessment concerning technology-related knowledge.									
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)							
			a	b	c	d	e	f	g	
	1.Attendance and participation	5%		✓		✓	✓	✓		
	2.Group presentation	10%	✓	✓	✓	✓	✓	✓		
	3.Mid-term test	20%	✓	✓	✓		✓	✓	✓	
	4.Individual Assignment	15%	✓	✓	✓		✓	✓	✓	
	5.Examination	50%	✓	✓	✓		✓	✓	✓	
	Total	100 %								
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Oral presentations are used to assess how well the students understand the content of materials and ability to organize and structure the materials. Individual assignments are used to test students' ability in understanding the materials and measure their evaluation abilities. Mid-term test and final examination are used to test students' overall ability in applying the knowledge learnt in the subject.									
Student Study Effort Expected	Class contact:									
	▪ Lectures/Seminars						39 Hrs.			
	Other student study effort:									
	▪ Reading subject textbooks and materials						42 Hrs.			

	Preparation for tutorial homework and presentation	27 Hrs.
	Total student study effort	108 Hrs.
Reading List and References	Andersen, D. C. (2017). Do Credit Constraints Favor Dirty Production? Theory and Plant-Level Evidence. <i>Journal of Environmental Economics and Management</i> 84, 189–208. Andersson, Julius J. (2019). "Carbon Taxes and CO2 Emissions: Sweden as a Case Study." <i>American Economic Journal: Economic Policy</i>, 11 (4): 1-30. Baumol, W.J., and Oates, W.E. (1988) <i>The Theory of Environmental Policy</i>. Cambridge University Press. Bovenberg, A Lans & Mooij, Ruud A. (1994). Environmental Levies and Distortionary Taxation, <i>American Economic Review</i>, 84(4), pp. 1085-1089, September. Bovenberg, A Lans & Goulder, Lawrence H. (1996). Optimal Environmental Taxation in the Presence of Other Taxes: General-Equilibrium Analyses, <i>American Economic Review</i>, 86(4), pp. 985-1000, September. Curtis, E. M. (2017). Who Loses Under Cap-And-Trade Programs? The Labor Market Effects of the NOx Budget Trading Program. <i>Review of Economics and Statistics</i> 100(1), 151–166. Dang, V. A., Gao, N., & Yu, T. (2022). Climate policy risk and corporate financial decisions: Evidence from the NOx budget trading program. <i>Management Science</i>. Fullerton, D., & Metcalf, G. E. (1997). Environmental Taxes and the Double-Dividend Hypothesis: Did You Really Expect Something for Nothing? (No. w6199). NBER. Grainger, Corbett A., and Charles D. Kolstad. (2010). Who Pays a Price on Carbon? <i>Environmental and Resource Economics</i>, 46(3), pp. 359-376. Greenstone, M. (2002). The Impacts of Environmental Regulations on Industrial Activity: Evidence from the 1970 and 1977 Clean Air Act Amendments and the Census of Manufactures. <i>Journal of Political Economy</i> 110(6), 1175–1219.	

	Muehlenbachs, L., E. Spiller, and C. Timmins. 2015. "The Housing Market Impacts of Shale Gas Development," <i>American Economic Review</i>, 105(12): 3633-59. Nguyen, J. H., & Phan, H. V. (2020). Carbon risk and corporate capital structure. <i>Journal of Corporate Finance</i>, 64, 101713. Phaneuf, D. J., & Requate, T. (2016). <i>A Course in Environmental Economics: Theory, Policy, and Practice</i>. Cambridge University Press. Vona, F., G. Marin, D. Consoli, and D. Popp (2018). Environmental Regulation and Green Skills: An Empirical Exploration. <i>Journal of the Association of Environmental and Resource Economists</i> 5(4),713–753. Weitzman, M.L. (1974). Prices vs. Quantities. <i>Review of Economic Studies</i>, 41(4), pp. 477-491. Yip, C. M. (2018). On the Labor Market Consequences of Environmental Taxes. <i>Journal of Environmental Economics and Management</i> 89, 136–152. Thomas, J. M. and Callan, S. J. (2013). <i>Environmental Economics and Management: Theory, Policy, and Applications</i>, 6e. Mason, OH: South-Western, Cengage Learning. Field, B. C., and Field, M. K. (2012). <i>Environmental Economics: An Introduction</i>, 6e. New York: McGraw-Hill. Booth, D.E. (1994). Ethics and the limits of environmental economics. <i>Ecological Economics</i> 9(3): 241-252. Johansson-Stenman, O. (1998). The importance of ethics in environmental economics with a focus on existence values. <i>Environmental and Resource Economics</i> 11(3-4): 429-442. Other academic journal articles.
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